CONGRESSIONAL APP CHALLENGE · APP DEMONSTRATION

# CareSphere

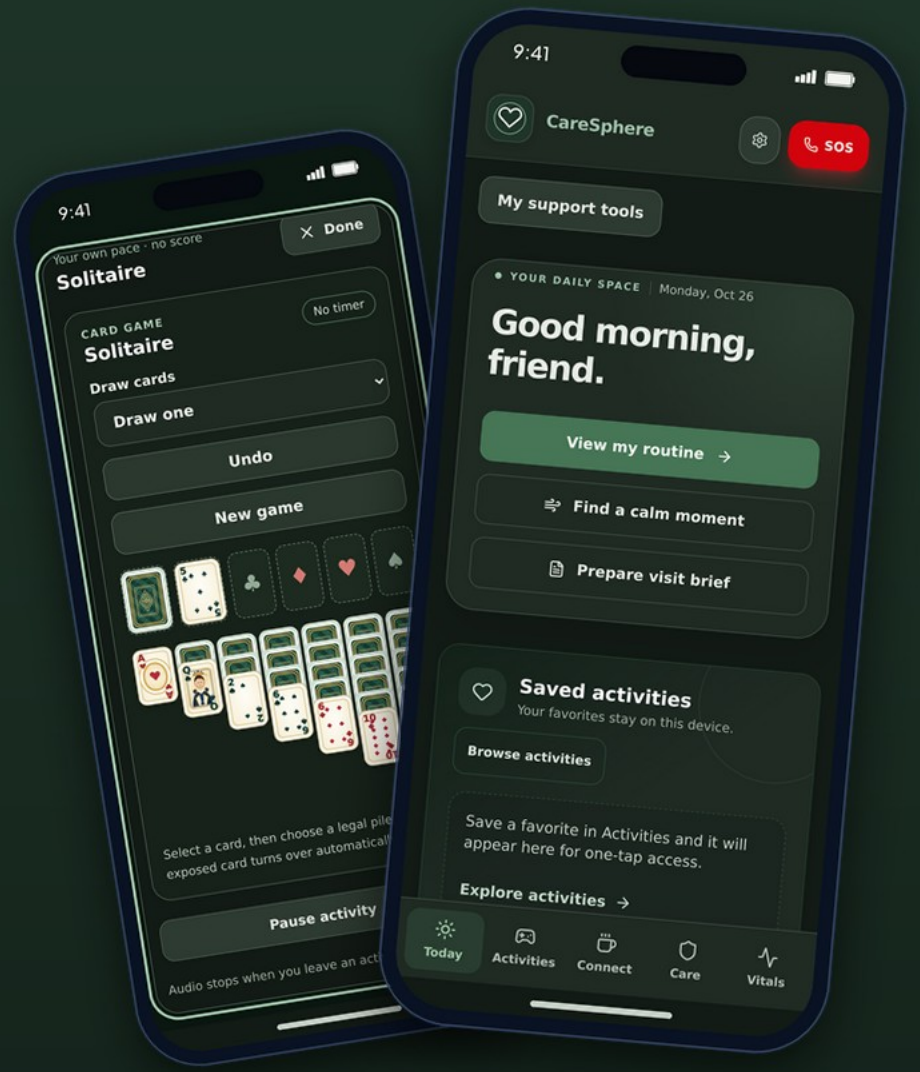
## Your wellness, your way.

Activities, daily routines, connection and help preparing for care visits.

DESIGNED FOR

Older adults, people living with dementia, autistic people, and the families and caregivers who support them.

Arnav Kochar · 9th grade · Congressional District 10



USER-FRIENDLY DESIGN

02

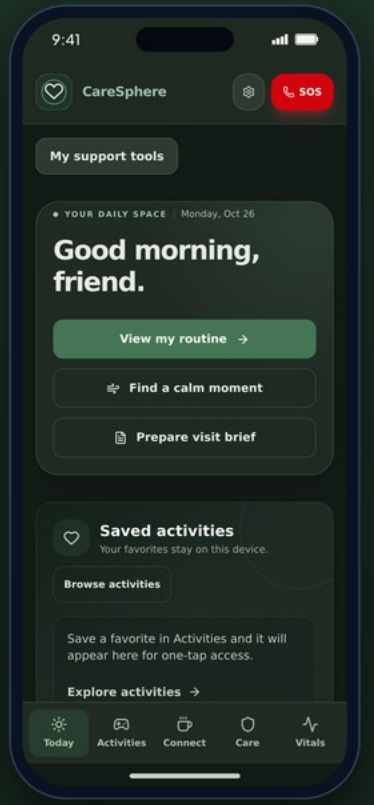
# Simple to use. Built around people.

## Clear choices. A comfortable pace.

### A predictable day
Mood check-ins, daily routines and medication reminders.

### A low-pressure activity
Familiar activities, optional sound and clear stop controls.

### A way to connect
Video rooms and a brief you prepare before a visit.



WHAT THE APP DOES

03

# Everyday support. Six connected areas.

## Sensory & interactive activities

16 activities with search, favorites and pace choices.

## Daily support

Mood check-ins, reminders and a moment to pause.

## Human connection

Coffee Circles and video calls you choose to join.

## Notes & care visits

Choose notes to share and prepare a visit summary.

## Health readings

Readings from a monitor or records you enter yourself.

## Make it your own

Simpler screens, language choices and comfort settings.

Also: emergency options · save a backup · decide what to share

CORE FEATURE · ACTIVITIES

04

# Activities. At your own pace.

## Solitaire




Draw · undo · no timer

## Word Search




Themes · word selection

## Bingo




Together on one device

## Picture Pairs




Reveal matching cards

Also: remembering patterns · picture stories · feelings · sorting · conversations · and more

05

# Research behind CareSphere.

Research informs the design; CareSphere itself has not been clinically evaluated.

| Population | Research findings | How CareSphere responds |
|---|---|---|
| **Dementia** | Digital interventions may support cognition and well-being (Ditton et al., 2025). | Memory games, stories, routines and calming tools. |
| **Autism** | Predictable structure and individual needs matter in technology design (Maun et al., 2024). | Consistent navigation, customizable settings and sensory tools. |
| **Older adults** | Video calls may improve social support and reduce loneliness (Noh et al., 2025). | Opt-in Coffee Circles, video calls and daily reminders. |

Personalized support · Accessible experiences · Meaningful connection

Sources: Ditton et al. (2025), Dementia; Maun et al. (2024), Journal of Autism and Developmental Disorders; Noh et al. (2025), Geriatric Nursing.

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LIVE APP DEMONSTRATION

# Let me show you CareSphere in action.

Choose what you need, in any order.

REAL APP · NO PRIVATE DATA

07

# Three platforms. Local-first care.

## Web

**JavaScript**

React · Vite · Web Audio  Web Bluetooth · Web Crypto

## iPhone source

**Swift**

SwiftUI · HealthKit  Optional Apple Foundation Models

## Android source

**Kotlin**

Android UI · BLE  Read-only Health Connect

## Encrypted local care data. Reviewed, selected sharing.

Jitsi uses external services. New native activity builds and physical-device validation remain pending.

WHO THIS COULD SUPPORT

08

# Different needs. Shared daily support.

## 61.2 million

### Older adults

Americans age 65+ · 2024

[S1] U.S. Census Bureau · 2024 data

## 7.4 million

### Alzheimer's dementia

Americans age 65+ · 2026 estimate

[S2] Alzheimer's Association · 2026

## 63 million

### Family caregivers

U.S. family caregivers · 2025

[S3] AARP / National Alliance for Caregiving · 2025

## 1 in 31

### Autism identification

8-year-olds in CDC monitoring communities · 2022

[S4] Centers for Disease Control and Prevention · 2022 data

Different years and populations. These figures overlap; they are not app-user counts.

09

# Reach someone. Keep control.

## Coffee Circles

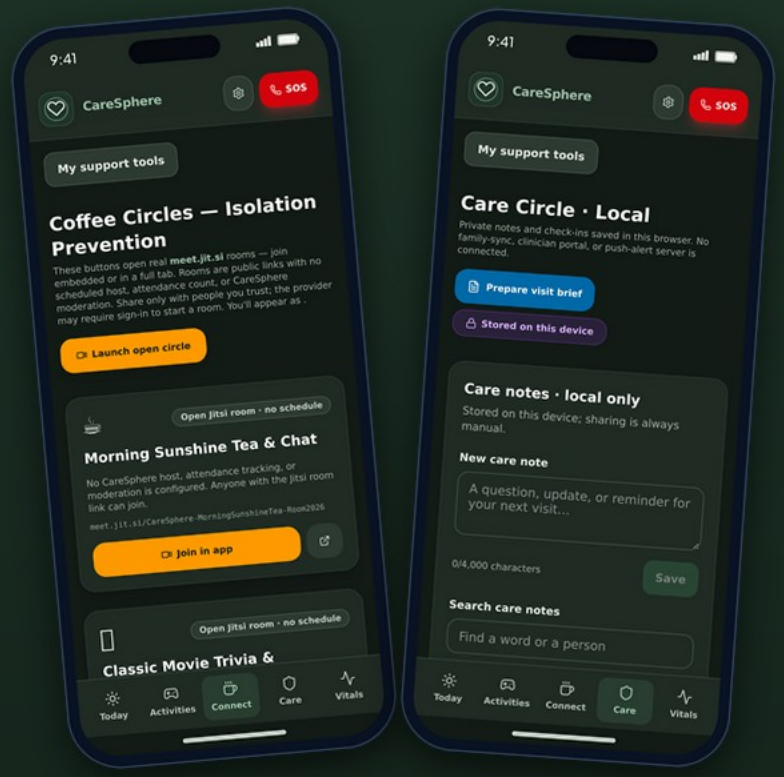
Choose a video room and join when you are ready.

## Care Circle

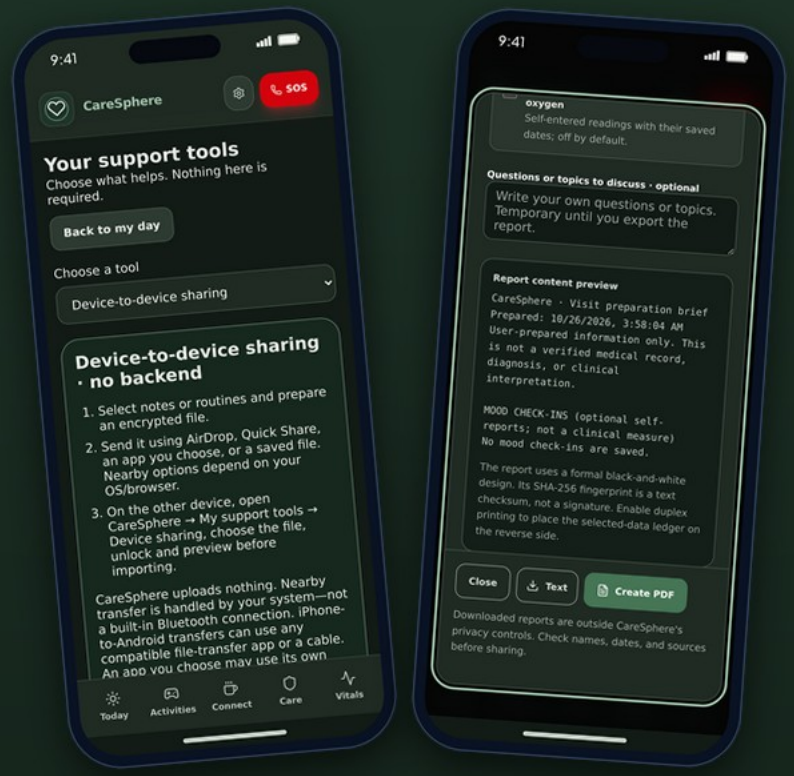
Keep notes on your device and prepare for conversations.

## No invisible sharing

Nothing is sent to family or a clinician on its own.



# Choose. Review. Then share.




## 01 · Select

Include only the records you choose.

## 02 · Review

Read your visit summary before saving it.

## 03 · Share intentionally

Review what you send. Nothing is shared on its own.

Keep saved reports safe. A visit summary is not a medical record.

WHAT COMES NEXT

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# Build with people. Keep improving.

| | |
|---|---|
| **Listen to people** | Ask people who may use the app what works for them. |
| **Check ease of use** | Find out which tasks feel clear and comfortable. |
| **Check on devices** | Test the latest activities on phones before release. |

**Next: listen, test and improve with the people who use it.**

Explore the project

# Care that stays yours.

Everyday support, on your terms.

EXPLORE THE PROJECT READ THE CODE EXPLORE THE APP

github.com/engineer2025-sudo/CareSphere